Md Mushfiqur Rahaman

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Academic Profile

Ph.D. student in Industrial Engineering with a concentration in Data Science at West Virginia University, with research interests in real time edge AI, model quantization, anomaly defect detection in additive manufacturing and AI-enabled smart manufacturing systems. Background includes an M.S. in Applied Mathematics and more than six years of industrial automation and Industry 4.0 experience in FMCG manufacturing at Unilever Bangladesh, Nestlé Bangladesh, and Unipolar Automation Technologies.

Education

- **West Virginia University** — *Aug. 2025–Present; Expected Aug. 2028*
  Ph.D. in Industrial Engineering, Concentration: Data Science — Morgantown, WV, USA
- **West Virginia University** — *Aug. 2025*
  Master of Science (M.S.) in Applied Mathematics — Morgantown, WV, USA
- **Chittagong University of Engineering and Technology** — *Mar. 2017*
  Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering — Chittagong, Bangladesh

Thesis

- **M.S. Thesis:** "Post-Training Quantization of Generative and Discriminative LSTM Text Classifiers: A Study of Calibration, Class Balance, and Robustness." West Virginia University, 2025. DOI: 10.33915/etd.13013.

Research Interests

- Edge AI and deployment of machine learning models on resource-constrained devices
- Post-training quantization, model compression, and efficient deep learning inference
- Computer vision for manufacturing inspection, additive manufacturing, and healthcare monitoring
- Industrial IoT, Industry 4.0, SCADA/PLC-based data acquisition, and smart manufacturing systems

Publications

- Rahaman, M. M., Chang, E., Haque, T., & Das, S. (2025). "Post-Training Quantization of Generative and Discriminative LSTM Text Classifiers: A Study of Calibration, Class Balance, and Robustness." *arXiv preprint arXiv:2507.09687.* DOI: 10.48550/arXiv.2507.09687.
- Chowdhury, M., Hossain, Q. D., Saha, P., & Rahaman, M. M. (2016). "Design, Fabrication and Performance Evaluation of a Three Electrode ECG Recorder." *Proceedings of the International Conference on Innovations in Science, Engineering and Technology (ICISET)*, pp. 1–4. DOI: 10.1109/ICISET.2016.7856500.
- Sammir, M. R., Alam, K. M. R., Saha, P., Rahaman, M. M., & Hossain, Q. D. (2018). "Design and Implementation of a Non-invasive Jaundice Detection and Total Serum Bilirubin Measurement System." *Proceedings of the International Conference on Electrical and Computer Engineering (ICECE)*, pp. 137–140. DOI: 10.1109/ICECE.2018.8636801.
- Sammir, M. R. et al. (2016). "Development of Tele-operated Robotic Hand Observing Movement of the Human Hand Using Gyro." *2016 7th International Conference on Mechanical Engineering and Renewable Energy.*

Acknowledged Research Contributions

- Acknowledged for helpful technical discussions and computing support in Haque, T., Syed, M. A. B., Jeong, B., Bai, X., Mohan, S., Paul, S., Ahmed, I., & Das, S. (2026). "Towards efficient real-time video motion transfer via generative time series modeling." *Multimedia Tools and Applications*, 85, Article 51. DOI: 10.1007/s11042-026-21138-4.

Ongoing Research

- **Edge deployment and quantization of foundation models for additive manufacturing inspection.** Working on domain adaptation and quantization of Segment Anything Model (SAM)-based segmentation models for XCT-based additive manufacturing defect and pore inspection, with deployment considerations for Qualcomm edge devices.

- **Post-training quantization of generative and discriminative LSTM text classifiers.** Studying quantization of neural network-based text classifiers for edge computing deployment using Brevitas, with focus on calibration quality, class balance, and robustness under low-bit quantization.

Research Methods and Tools

- **Deep Learning and Generative Modeling:** PyTorch, LSTM, GRU, VRNN, SRNN, Transformer models.
- **Model Compression and Edge AI:** Post-training quantization, Brevitas, ONNX, Qualcomm QNN/SNPE, low-bit inference, edge deployment on resource-constrained devices.
- **Computer Vision and Segmentation:** Segment Anything Model (SAM), Conv-LoRA adaptation, XCT image segmentation.
- **Industrial AI and Time-Series Analytics:** Sensor forecasting, anomaly detection, logistic-regression-based classification, SCADA/PLC/OPC/SQL-based manufacturing data pipelines.
- **Research Computing:** Kubernetes, Nautilus/NRP, Linux servers, GPU training, Raspberry Pi, Qualcomm RB3 Gen2.

Selected Academic and Research Projects

- **Non-invasive Hemoglobin Measurement System with Image Processing.** Developed a handheld goggle prototype with an integrated camera to capture conjunctiva images and estimate blood hemoglobin level using an image-processing-based correlation model. The prototype was evaluated using data collected from thirty participants and achieved approximately 94% detection accuracy.

- **Industrial Data Acquisition and Reporting Systems.** Designed and implemented automation-data pipelines involving OPC servers, SQL databases, SCADA systems, and reporting dashboards for manufacturing lines and utilities, providing practical background for current research in smart manufacturing and industrial AI.

Professional and Industrial Experience

- **Unilever Bangladesh Ltd.** — *Mar. 2020–Jul. 2023*
  Manager–Electrical and Automation — Chittagong, Bangladesh
  - Led Industry 4.0 and digital factory implementation activities, including plant-level automation, electrical systems, utility monitoring, and cost optimization.
  - Served as country-level Technical Assistance/Subject Matter Expert for automation, instrumentation, electrical safety, and electrical system design.
  - Implemented energy monitoring and reporting systems, Central Automation System (CAS) network backbone, OPC/Azure-based Digital Factory Operating System (DFOS), SQL/SSRS-based statistical process control dashboards, and SCADA/SQL-based utility monitoring.
  - Coordinated CAPEX and engineering projects, including PCC panel replacement, electrical safety upgrades, and commissioning of two European bottle-filling and packing lines during the COVID lockdown without OEM presence.
  - Awarded "Best Engineer" in 2021 and received the Supply Chain Director's Award in 2022 for talent development in control and instrumentation.

- **Nestlé Bangladesh Ltd.** — *Dec. 2018–Feb. 2020*
  Officer–Electrical and Automation — Gazipur, Bangladesh

- Maintained electrical and automation systems in a noodle manufacturing plant and supported spare-parts planning for manufacturing lines.
- Commissioned FreeWeigh.Net Server (Mettler Toledo) for real-time weighing data collection and analysis, WIPOTEC-OCS checkweigher, TOPAC vertical filling and wrapping machine, and automation improvements for the oil cooling system.

- **Unipolar Automation Technologies** — *Mar. 2017–Dec. 2018*
  Project Engineer — Dhaka, Bangladesh
  - Designed industrial control and instrumentation systems, prepared bills of materials, and performed installation and commissioning for automation projects.
  - Led or contributed to PLC lab setup and training at CUET, resin manufacturing plant automation at Berger Paints Bangladesh, 132 kV/33 kV substation SCADA implementation at KYCR Steel Mill, substation ACB interlock design at Nestlé Sreepur Plant, and pump station automation for Dhaka WASA.

Additional Technical and Industrial Skills

Programming and Data Science: Python, MATLAB, C, Java, SQL, Oracle, JMP
Machine Learning and Edge AI: PyTorch, model quantization, anomaly detection, computer vision, Kubernetes
Industrial Automation: PLC/HMI/SCADA systems; Allen-Bradley, Siemens, Schneider; VFD and servo systems
Industrial IoT and Communication: OPC/Kepware, Azure basics, Ethernet, Modbus, PROFINET, Cisco Firewall basics
Servers and Systems: Linux Server, Windows Server, VMware Server, Proxmox, Ubuntu
Edge and Embedded Devices: Qualcomm RB3 Gen2, Raspberry Pi, Arduino, PIC microcontroller, robotics environment
Engineering Tools: AutoCAD, Proteus (ISIS and ARES), P-Spice, Cadence EDA Tool, SAP

Public Speaking and Seminar

- **Keynote Speaker, "Seminar on Landing on Industry with Future-fit Skillset"** — *Dec. 8, 2022*
  Organized by the Department of Electrical and Electronic Engineering, CUET, on the occasion of EEE Day
- **Keynote Speaker, "Seminar on IoT in Industrial Automation–The Journey towards Industry 4.0"** — *Dec. 2021*
  Organized by the Robo Mechatronics Association, CUET
- **Guest Speaker, "An Open Talk on Science and Technology"** — *Feb. 20, 2017*
  Organized by former students of Chittagong University Laboratory School and College

Leadership and Student Organization

- **Co-Chair / Vice President** — *Dec. 2015–Dec. 2016*
  IEEE Student Branch, CUET — Chittagong, Bangladesh
- **Organizing Secretary** — *Oct. 2015–Dec. 2016*
  Robo Mechatronics Association, Bangladesh — Chittagong, Bangladesh
- **President** — *Sep. 2015–Dec. 2016*
  Greater Barisal Student Association, CUET — Chittagong, Bangladesh

Selected Embedded Systems and Robotics Projects

- **Robotics:** Bi-pedal robot using a six-DOF servo based model; quad-rotor robot with GPS-based location tracking; obstacle-detecting robot; maze-solving robot; remote-controlled robot; hovercraft with brushless DC motor.
- **Automation:** Electronic voting machine with biometric fingerprint and RFID; sunlight-tracking solar panel; RFID-based library automation; automatic car glass wiper; GSM-based automatic electric billing system; lift automation; microcontroller-based car parking automation.
- **Biomedical and Communication Systems:** Heartbeat monitoring system using infrared technology; brain-computer interfacing with NeuroSky EEG headset; geosynchronous satellite receiving station; RF-based railway station detection and train display system.